

Florian FRANZEN

Architecture | Infrastructure | Research | Leadership

@ Florian@Franzen.io
github.com/FlorianFranzen
scholar.google.com/OiBuDrsAAAAJ
 Zug, Switzerland

florian.franzen.io
linkedin.com/in/FlorianFranzen



After years in neuroscience and academia, I transitioned into decentralized protocol engineering with a focus on blockchain interoperability. At the Web3 Foundation, I co-developed the implementable Polkadot specification and shepherded multi-client standardization. Since then, I have provided technical leadership across Substrate-, Ethereum-, and Solana-based projects, covering protocol research, infrastructure engineering, and deployment operations. Currently I consult multiple startups in the Blockchain space and operate my own entity, F.R.E.D..

I bring extensive expertise spanning hardware and software development, cloud and bare-metal infrastructure, data analysis and machine learning (including AI/LLMs), full-stack web development, embedded systems, security engineering, reverse engineering, and prototype manufacturing.

SKILLS

Languages	Rust (expert), C/C++ (expert), Go (advanced), Haskell (proficient), Python (advanced), TypeScript (proficient), SQL
Protocols	Polkadot (Substrate, PolkaVM), Solana (Anchor), Ethereum (Solidity, Hardhat)
Infrastructure	Ansible, Docker/Podman, Kubernetes/Helm, Nix/NixOS, CI/CD, Networking, Linux
Tooling	Git, CMake, Ghidra, IDA Pro, Fuzzing, AsciiDoc, Doxygen, $\LaTeX$
Hardware	embedded development, PCB design (KiCad), 3D printing/CAD, electronics repair
Spoken Languages	English (fluent), German (native)

FREELANCE EXPERIENCE

Current 2023	Chief Engineer, FRANZEN RESEARCH, ENGINEERING & DEVELOPMENT (F.R.E.D.) Hardware and software prototyping, embedded development, scientific computing, data digitization, electronics repair, blockchain validator operations, and infrastructure management. C/C++ CAN CXADC ESP32 KiCad Nix Ollama OpenSCAD ROCm RT-Linux Rust Smartcard VHS-Decode
Current	Research Consultant, FLOWRA Lead technical research, inform product strategy, support investor due diligence, review code performance, author and publish technical articles. Alpenglow Jito MEV Rust Solana Tokio
Current	Integration Consultant, STEALTH STARTUP Advise on protocol architecture, mainnet integration strategy, and genesis launch execution. Intents MEV-protection Quantum-safety Rust Sharding
2026 2025	Senior Rust Engineer, RAIKU LABS LTD Design and implement auction marketplace. Interface with on-chain programs and data. Optimize performance (latency, jitter), scheduling and blockspace utilization. Support DevOps with upcoming changes. Liaison for the Research team. Anchor Helm Jito Kubernetes MEV Rust Solana Tokio Tonic
2025	Product and DevOps Engineer, ANALOG Design, implement, and deploy the official Substrate-Ethereum bridge. Containerize infrastructure for Kubernetes/Helm deployments, optimize reliability and cost, manage on-chain runtime upgrades and other governance. Ansible Docker Helm Kubernetes Linux Podman React Rust Solidity Substrate TypeScript WASM
2019 2018	Software and Network Engineer, WHITE MATTER LLC Port a C++ API from Winsock to cross-platform Asio. Design binary network protocols, optimize packet throughput, latency and jitter, enforce thread safety, implement unit tests, and extend the OpenEphys plugin. Asio C++17 Catch2 CMake Doxygen fmtlib JUCE Linux spdlog Windows
2015	Software Engineer, BLACKROCK MICROSYSTEMS LLC Add Blackrock file import and live data streaming to Neuroscope. Extend the Cerebus SDK, refactor and debug Neuroscope core, and establish CI/CD pipelines. C++ CMake Qt Blackrock SDK Linux Windows

PROFESSIONAL EXPERIENCE

Dec. 2025 Aug. 2025	Head of Protocol Research, GLUWA, Remote Direct research strategy and execution. Support product goals across technical teams. Lead evaluation of emerging technologies and advise the Chief Architect and Head of Engineering on critical technical decisions. Cairo Python Rust Solidity Substrate TypeScript
Aug. 2025 Dec. 2024	Head of Infrastructure, ANALOG, Remote Oversee core blockchain infrastructure: mainnet launch, feature development, DevOps/SecOps, exchange and wallet integrations. Execute on-chain governance, fund management and runtime upgrades alongside founders. Alloy GraphQL Hardhat Helm Kubernetes PolkaVM React Rust Substrate TypeScript Tokenomics WASM
Dec. 2024 Jun. 2023	Head of Research, ANALOG, Remote Research and prototype core protocol components. Lead DevOps/SecOps and the technical side of product management, hiring and partner integrations. Report directly to founders. Cryptography Game Theory Helm Kubernetes Protocol Design Rust Security Substrate Tokenomics WASM
May. 2023 Mar. 2020	Polkadot Specification Lead, WEB3 FOUNDATION, Zug, Switzerland Co-develop the implementable Polkadot specification alongside the research team. Validate external implementations through a custom cross-implementation test suite. Shepherd the multi-client standardization process. AsciiDoc C++ CMake Go Gossamer Kagome Julia k-framework Python Ruby Rust Substrate WASM
May. 2019 Jul. 2016	Doctoral Researcher in Neuroinformatics, RESEARCH CENTER CAESAR, Bonn, Germany Develop 3-D environment and pose reconstruction from video data using non-linear optimization of multi-view geometric graphs. Build an ultra-lightweight tetherless recording platform for GPS, IMU, and barometric sensors. Reverse-engineer and extend a consumer-grade 360° camera. Develop advanced camera calibration, synchronization, and spatial tracking system. Blender C C++ CAN CMake Docker IDA Pro i²c Jupyter Matlab Nix OpenCV Python LSD-SLAM SPI Teensy TizenRT Qt
Jul. 2016 Nov. 2015	Network & System Administrator, RESEARCH CENTER CAESAR, Bonn, Germany Install and configure wired and wireless infrastructure for 200+ clients, including firewalls, routing, switching, VLANs, 802.1x authentication, VPN, RADIUS, and SIP telephony. Bash Cisco ASA Cisco IOS Cisco ISE Cisco WLC Microsoft Active Directory

EDUCATION

2014	Master of Neural & Behavioural Science, INTERNATIONAL MAX PLANCK RESEARCH SCHOOL, Tübingen, Germany Thesis: "Evaluation of a Statistical Model for Cortical State Identification" Final grade: 2.1
2012	Bachelor of Cognitive Science, UNIVERSITY OF TÜBINGEN, Tübingen, Germany Thesis: "Neural Circuits of Locomotor-related Response Modulation in Mouse Primary Visual Cortex" Final grade: 2.1

PUBLICATIONS

- 2014 P. Putzky, **F. Franzen**, G. Bassetto, J. H. Macke. A Bayesian model for identifying hierarchically organised states in neural population activity. *Advances in Neural Information Processing Systems* 27. 2014.
- 2013 A. Vaiceliunaite, S. Erisken, **F. Franzen**, S. Katzner, L. Busse. Spatial integration in mouse primary visual cortex. *J Neurophysiol.* 2013 May 29.

TEACHING

2023 10 days	Polkadot Blockchain Academy, UNIVERSIDAD DE BUENOS AIRES, Buenos Aires, Argentina Teaching cryptography, economics and blockchain
2022 1 week	Polkadot Blockchain Academy, CORPUS CHRISTI COLLEGE, Cambridge, England Teaching economics through interactive games
2015 2 weeks	Open Lab Ware Course, TREND IN AFRICA, Addis Ababa, Ethiopia Building and calibrating delta-style 3D printers, CAD, electrical circuit design, microcontroller programming